

Iris Dataset — Exploratory Data Analysis

Findings Report | AI/ML Engineering Internship | June 2026

1. Dataset Overview

The Iris dataset, introduced by statistician Ronald Fisher in 1936, contains measurements of 150 iris flowers across three species: **Setosa**, **Versicolor**, and **Virginica**. Each sample has four features (sepal length, sepal width, petal length, petal width) measured in centimetres. The dataset is **perfectly balanced** (50 samples per class) and contains **no missing values**, making it ideal for classification EDA.

Feature	Mean	Std Dev	Min	Max
Sepal Length (cm)	5.84	0.83	4.30	7.90
Sepal Width (cm)	3.06	0.44	2.00	4.40
Petal Length (cm)	3.76	1.77	1.00	6.90
Petal Width (cm)	1.20	0.76	0.10	2.50

2. Key Findings

Finding 1 — Setosa is Perfectly Linearly Separable

Iris Setosa is completely distinguishable from Versicolor and Virginica using petal measurements alone. Its petals are dramatically smaller (mean petal length 1.46 cm vs 4.26 cm for Versicolor and 5.55 cm for Virginica). This makes it trivially classifiable with even a simple threshold rule, e.g. petal length < 2.5 cm → Setosa.

Finding 2 — Petal Features Are the Strongest Discriminators

The petal-based features provide far better species separation than sepal features:

- **Petal Length**: largest variance across species — best single feature for classification
- **Petal Width**: near-perfectly correlated with petal length ($r = 0.96$)
- **Sepal Width**: weakest discriminator — overlaps across all three species

Finding 3 — High Multicollinearity Among Petal Features

The correlation heatmap reveals that petal length and petal width are nearly redundant ($r = 0.96$). Both are also strongly correlated with sepal length ($r = 0.87$ and 0.82 respectively). This suggests that a single petal dimension may suffice for a simple classifier, while PCA could reduce the feature space to 2 components for ML models.

Finding 4 — Versicolor and Virginica Partially Overlap

Unlike Setosa, Versicolor and Virginica share overlapping petal and sepal measurement ranges. Their petal length ranges overlap between approximately 4.5–5.1 cm. This partial overlap means a non-linear classifier (e.g. SVM with RBF kernel, Random Forest) will outperform a simple linear classifier on this boundary.

Finding 5 — Sepal Width Shows Reverse Pattern

Unusually, Setosa has the *widest* sepals (mean 3.43 cm) while Versicolor has the narrowest (mean 2.77 cm). This is the opposite pattern from every other feature. Sepal width is also weakly negatively correlated with petal features ($r = -0.37$ to -0.43), making it a less reliable standalone predictor but still useful in combination.

3. Species Mean Comparison

Feature	Setosa	Versicolor	Virginica
Sepal Length	5.01 cm	5.94 cm	6.59 cm
Sepal Width	3.43 cm	2.77 cm	2.97 cm
Petal Length	1.46 cm	4.26 cm	5.55 cm
Petal Width	0.25 cm	1.33 cm	2.03 cm
Petal Area*	0.37 cm ²	5.72 cm ²	11.30 cm ²

*Petal Area = Petal Length × Petal Width (engineered feature)

4. Visualizations

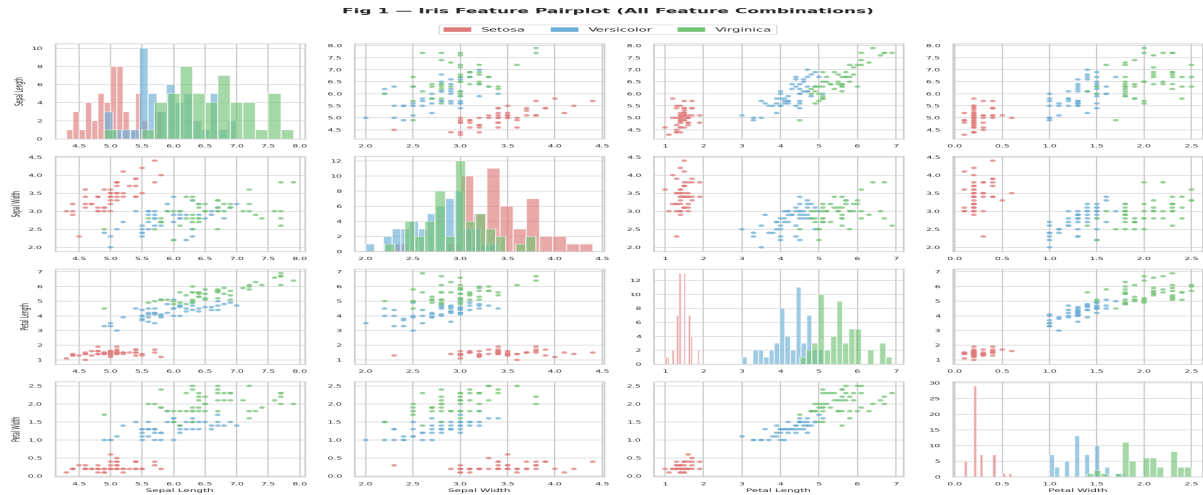


Fig 1: Feature Pairplot — All 4 Features vs All 4 Features

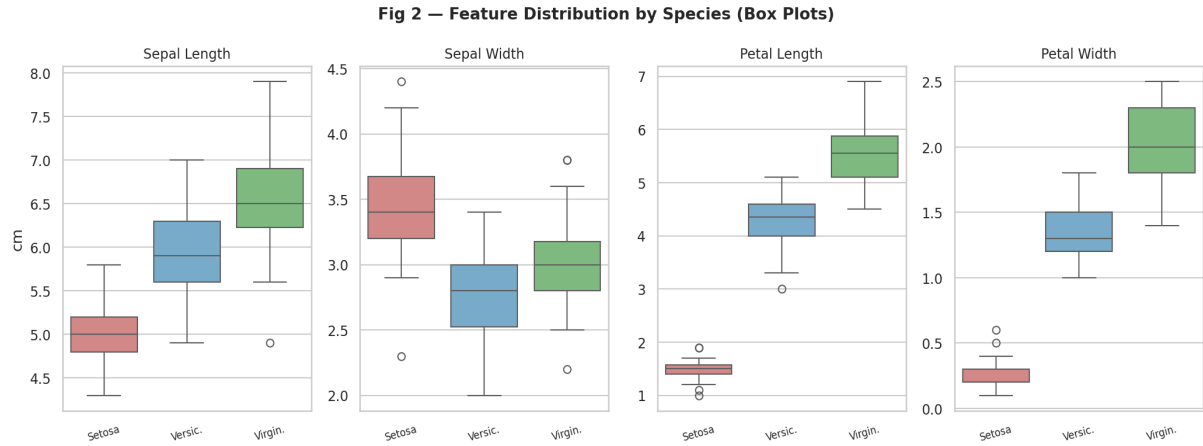


Fig 2: Box Plots — Feature Spread and Outliers by Species

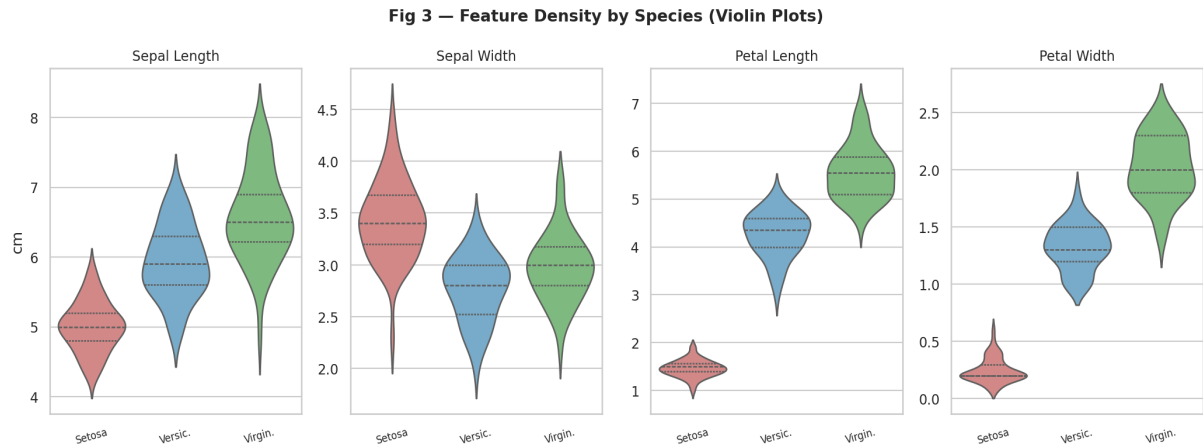


Fig 3: Violin Plots — Full Distribution Density by Species

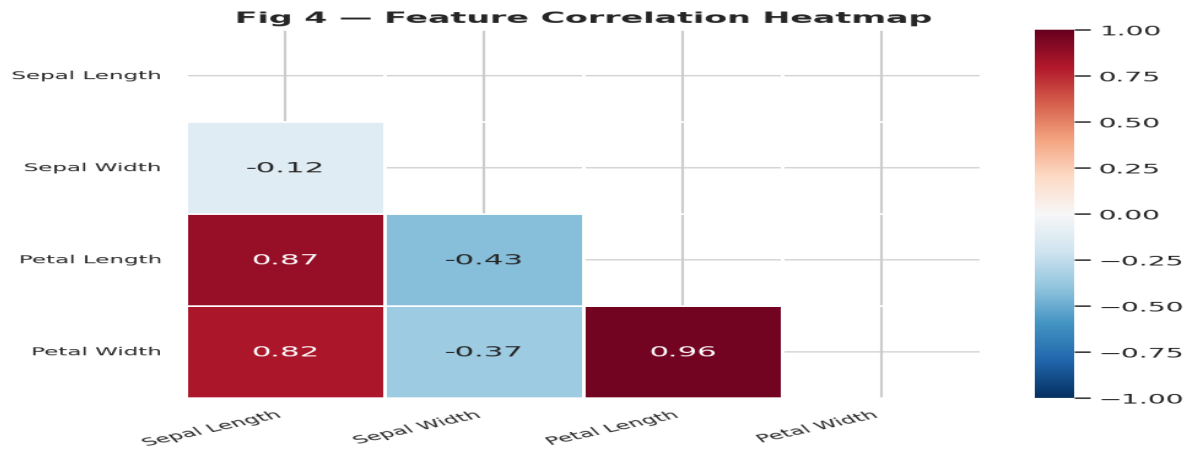


Fig 4: Feature Correlation Heatmap

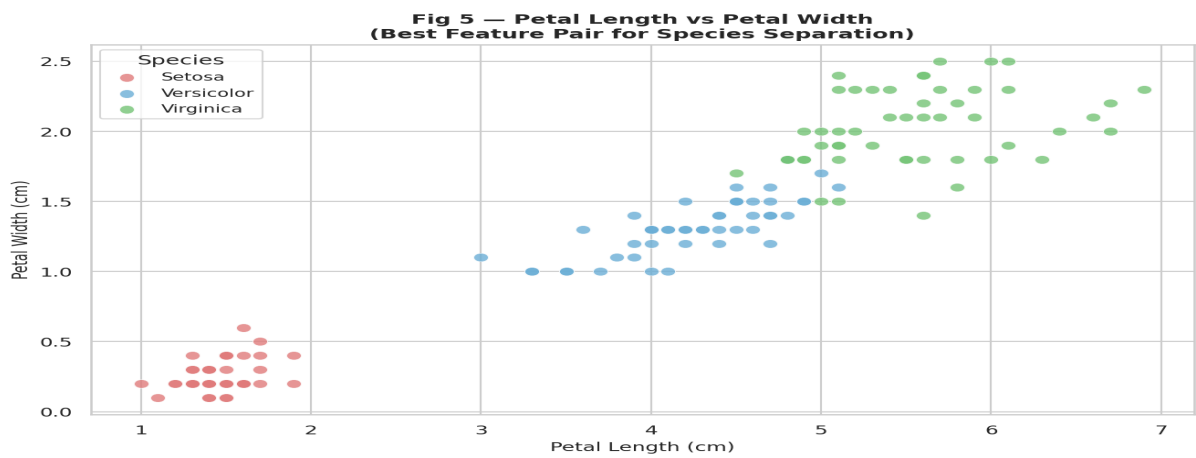


Fig 5: Petal Length vs Petal Width — Best Separating Feature Pair

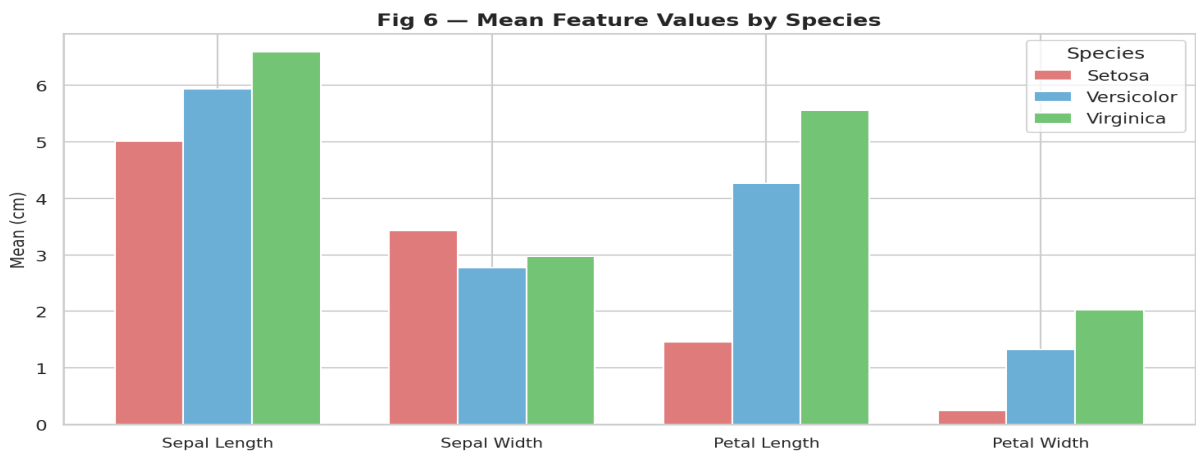


Fig 6: Mean Feature Values Comparison Across Species

5. ML Recommendations

- Use **Petal Length** or **Petal Width** as primary features — they explain the most variance.
- Apply **PCA**: 2 components will retain ~97.8% of variance due to high feature correlation.
- For Setosa: any simple rule (petal length < 2.5 cm) achieves 100% accuracy.
- For Versicolor vs Virginica: use SVM (RBF kernel) or Random Forest for the overlapping boundary.
- Baseline target: >95% accuracy is achievable with Logistic Regression; >97% with SVM.
- Feature selection: consider dropping Sepal Width or using it only in ensemble models.

